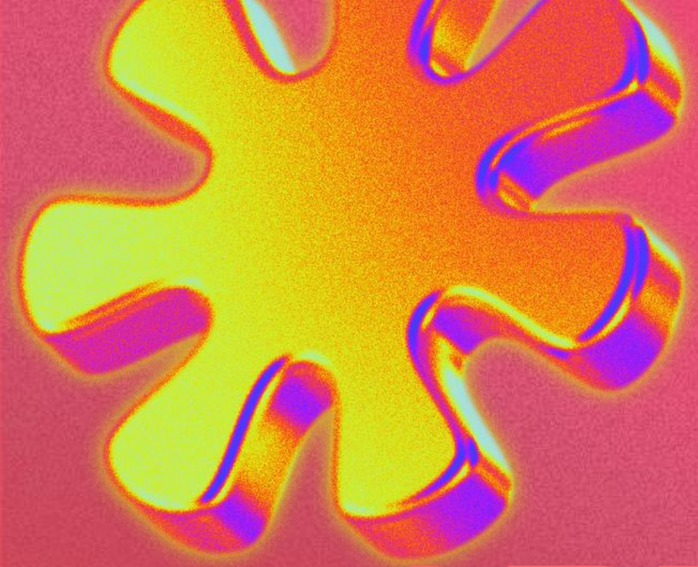
SPECTROSCOPY

A. Thoms

J. Raugh & S. Jon







Objective

Is to...

- Observe and measure the spectrum of gas-discharge lamps using...
  - Diffraction grating and Prism
  - Comparison of experimental vs. expected
  - Calculation of constants (Rydberg constant & Index of Refraction)

Background Physics

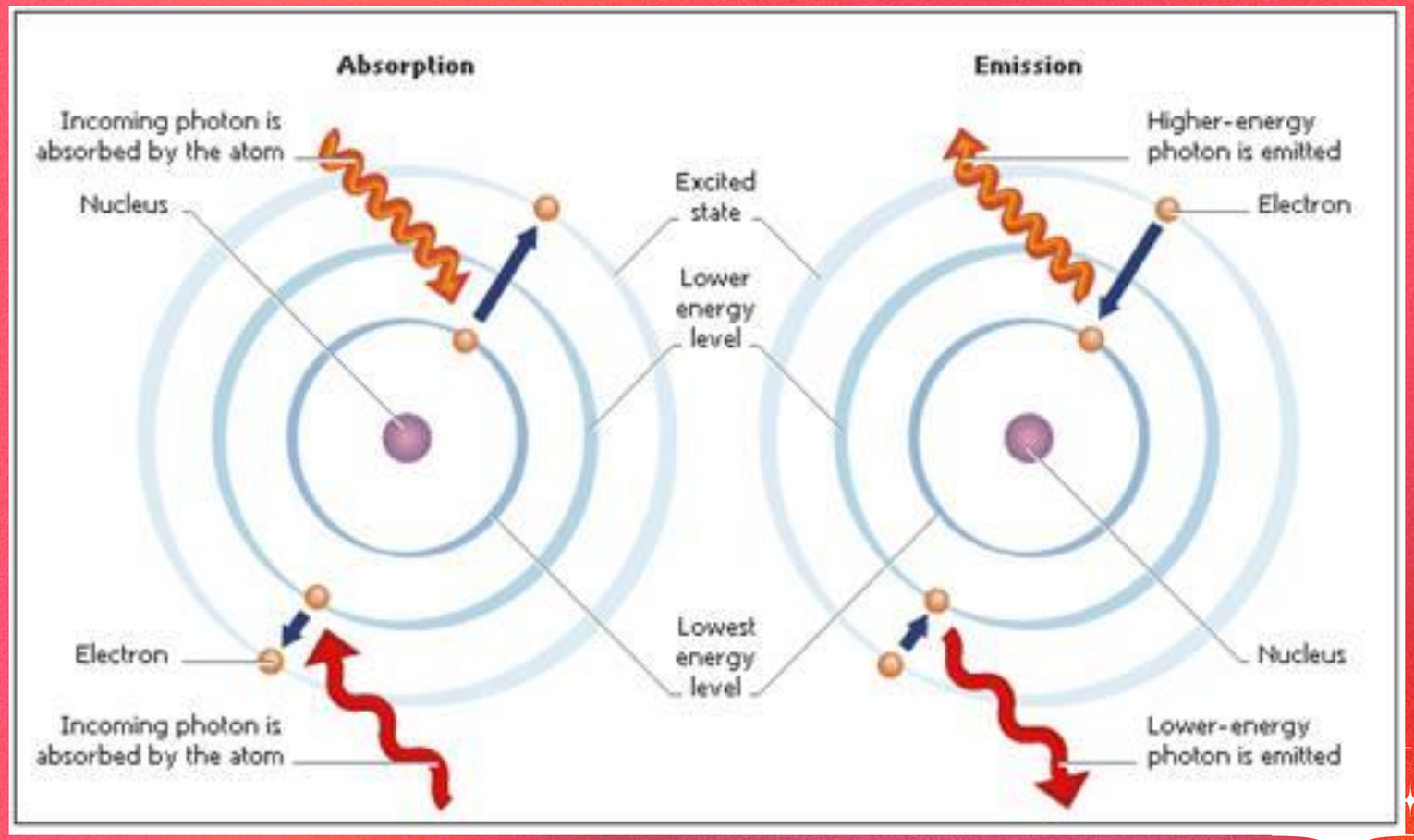
What is Spectroscopy?

Spectroscopy is the study of the behavior of light and how different elements interact with light.



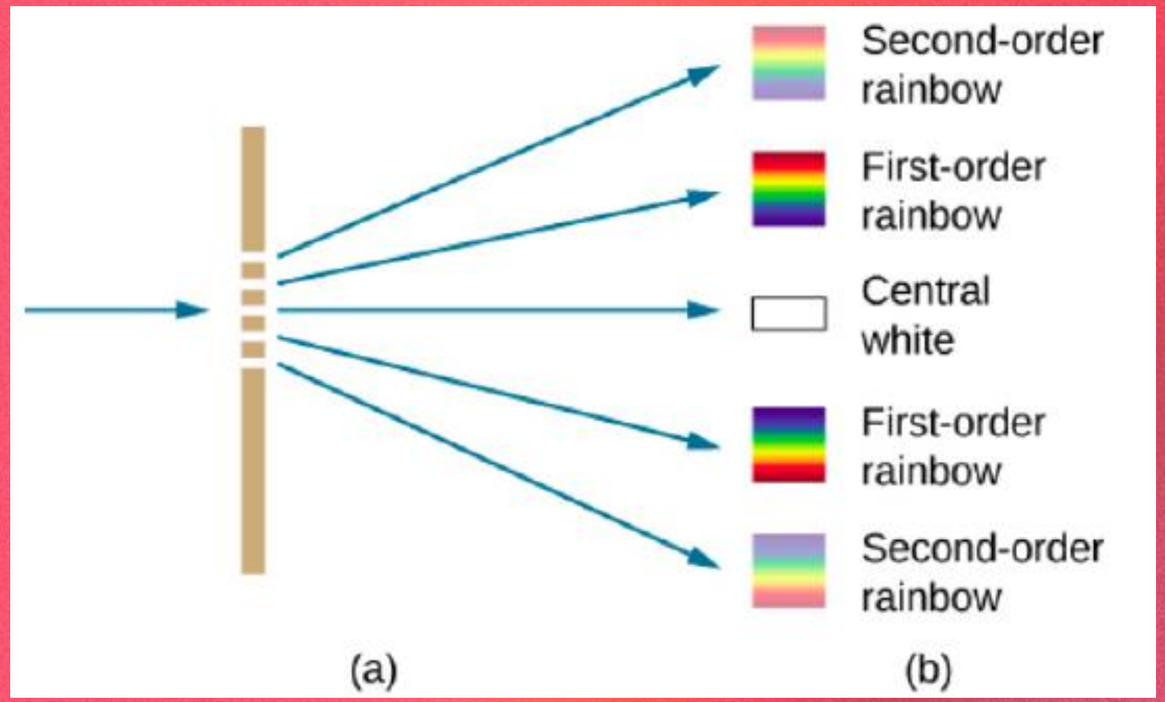
Basic terminology

- **Dispersion:** separation of white light to obtain a color spectrum
- **Diffraction:** when waves bend around an obstacle
- **Refraction:** the change in the direction of light when traveling through one medium to another



- **Emission spectrum** -> unique combination of spectra lines when a material is exposed to heat or electricity.
- This is due to the difference in **absorption** or **emission** rates of materials, caused by the various combinations of photons and electrons possible.

Diffraction Grating



Wavelength

Measurable quantity of light as a wave:

λ = Wavelength (nm)

d = Distance between lines

θ = Angle of Deviation

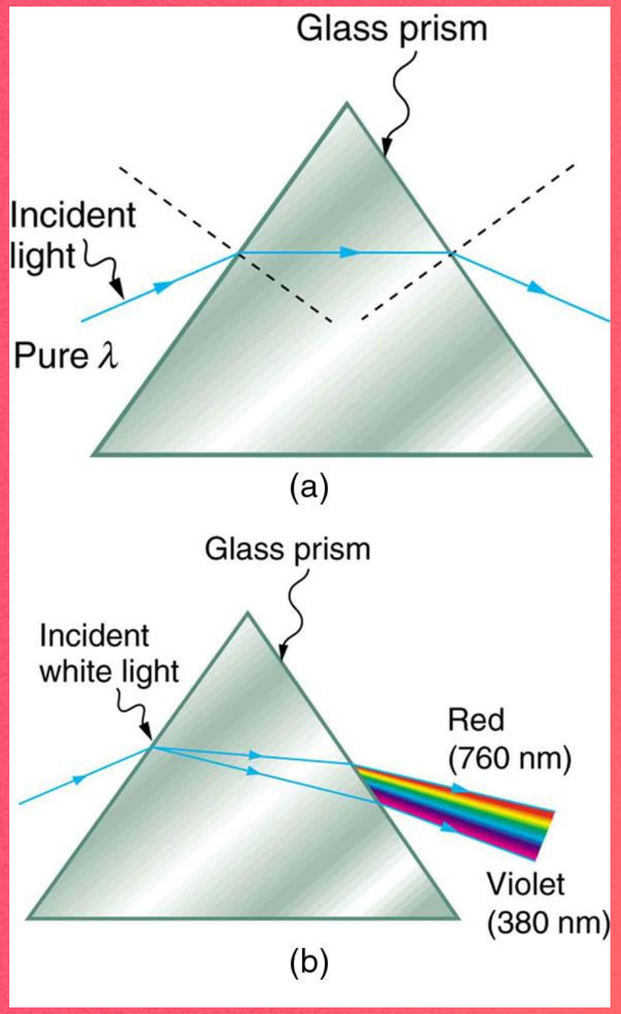
m = Order

$$\lambda = (d \sin\theta)/m$$

4.5: Diffraction Gratings - Physics LibreTexts

- Grooves are distanced equally apart
  - (slit/lines per ___)
- Transmission of light -> sharp patterns from constructive interference

Prism



Index of Refraction

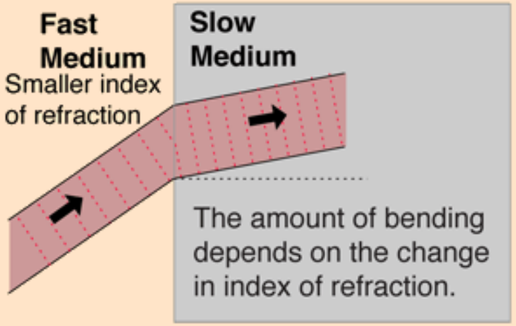
"Density" -> light traveling through a medium:

n = Index of Refraction

α = Vertex Angle

δ_{min} = Angle of Minimum Deviation

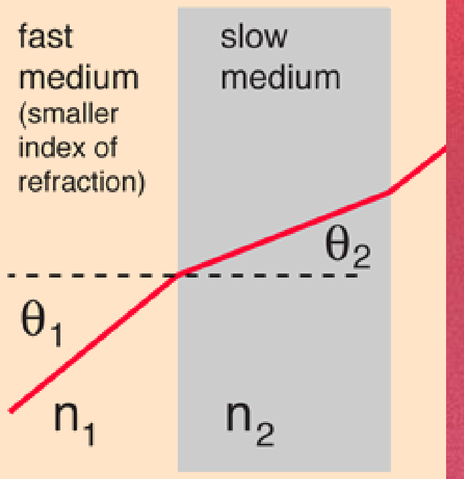
$$n = (\sin(\alpha/2 + \delta_{min}/2))/\sin(\alpha/2)$$



index of refraction $n = \frac{c}{v}$ *velocity of light in vacuum* / *velocity of light in the medium*

Snell's Law

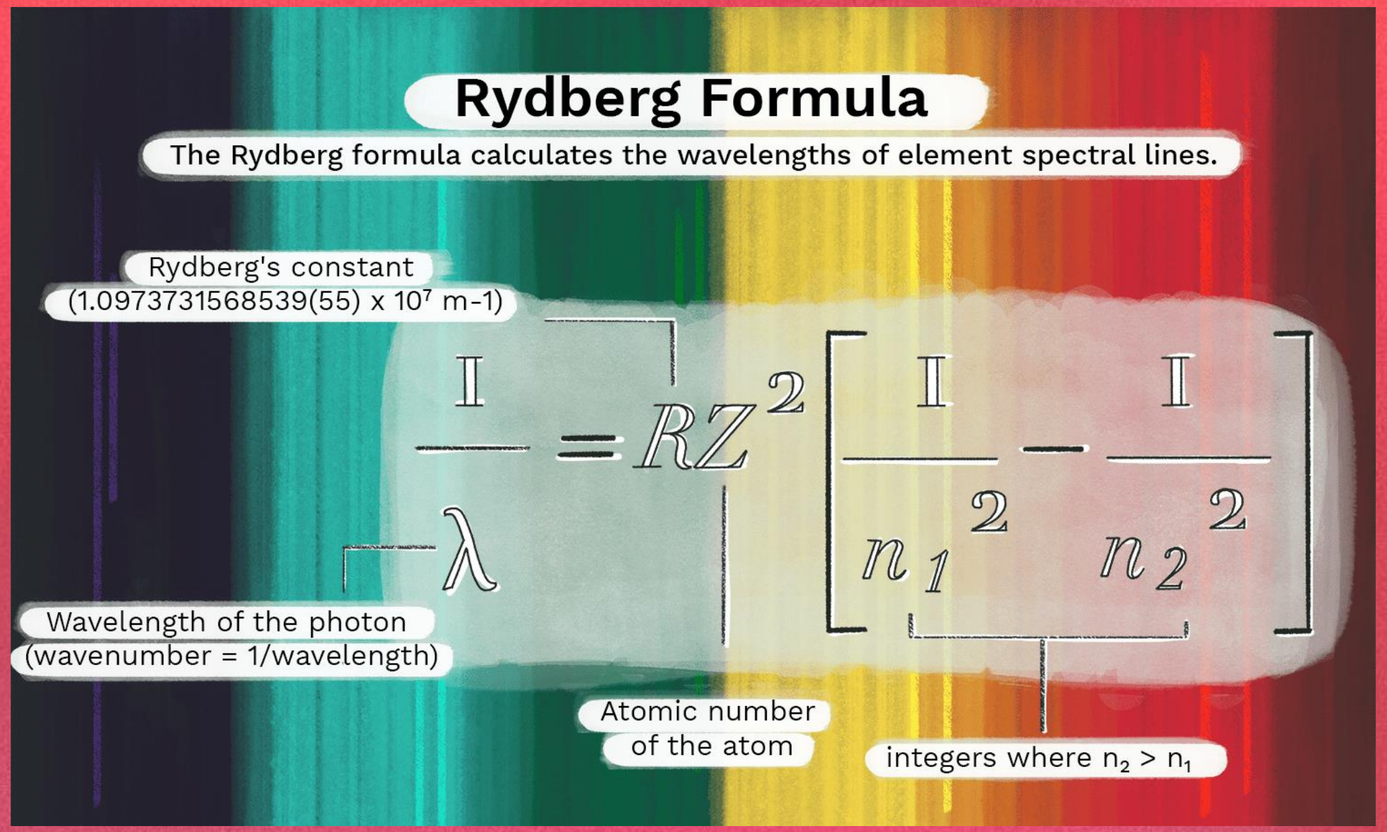
$$\frac{n_1}{n_2} = \frac{\sin\theta_2}{\sin\theta_1}$$



[25.5: Dispersion - Rainbows and Prisms - Physics LibreTexts]

[Refraction of Light (gsu.edu)]

Rydberg Constant



[What Is the Rydberg Formula and How Does It Work? (thoughtco.com)]

Rydberg Constant

Constant in the Rydberg formula that produces any of the lines in the Hydrogen spectrum:

λ = Wavelength (nm)

R_H = Rydberg Constant

m= Balmer Series

$1/\lambda = R_H((1/4)-(1/n^2))$

History

Evolution of classical spectroscopy...



Isaac Newton

1704 "*Opticks*" -> Prism Experiment



William Herschel

1800 - > Infrared Spectrum



Joseph von Fraunhofer

1814 - > Fraunhofer lines

Procedure

A

Goals:

- Compare experimental values of wavelength with expected
- Calculate the Rydberg Constant for Hydrogen and Deuterium
- Calculate a (lines) of the unknown diffraction grating

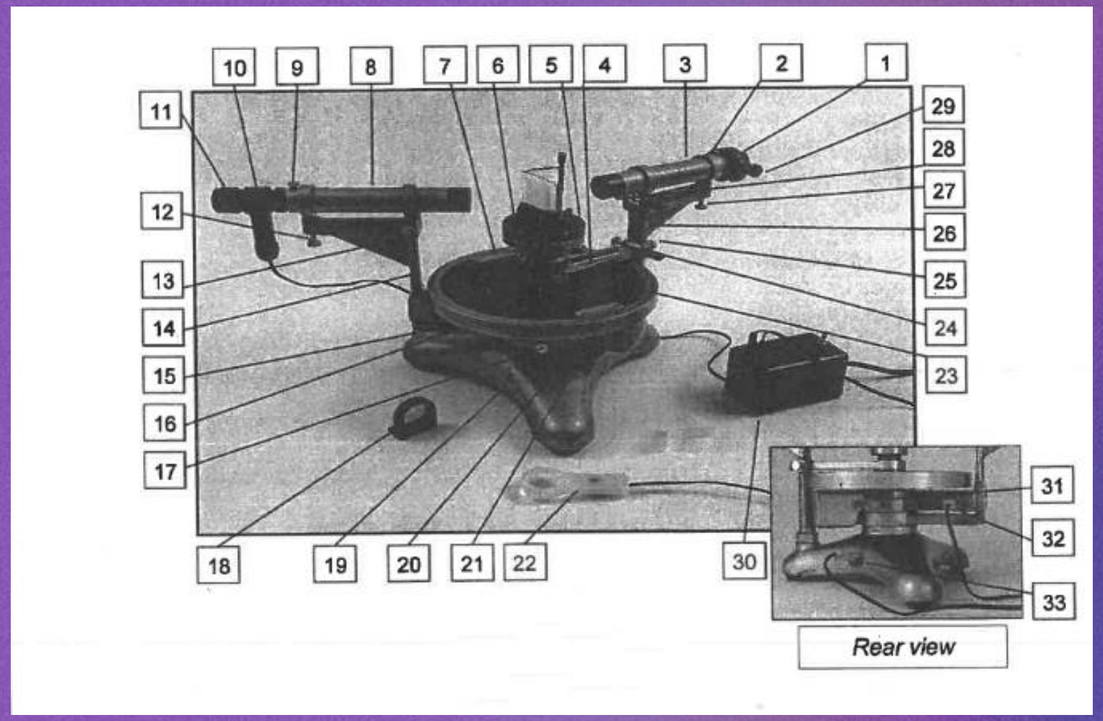
B

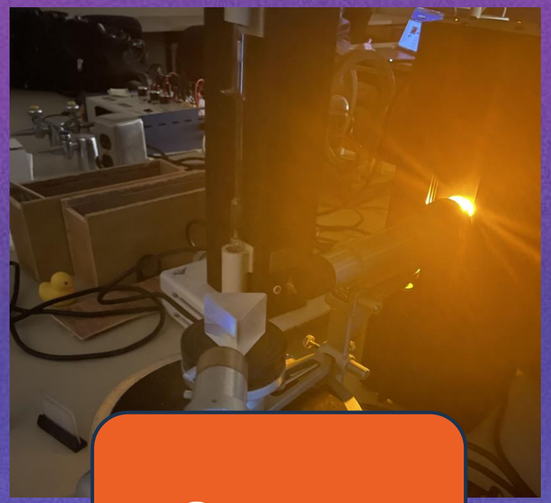
Goals:

- Plot the angle of deviation vs. wavelength
- Calculate the Rydberg Constant for Hydrogen
- Produce the Index of Refraction

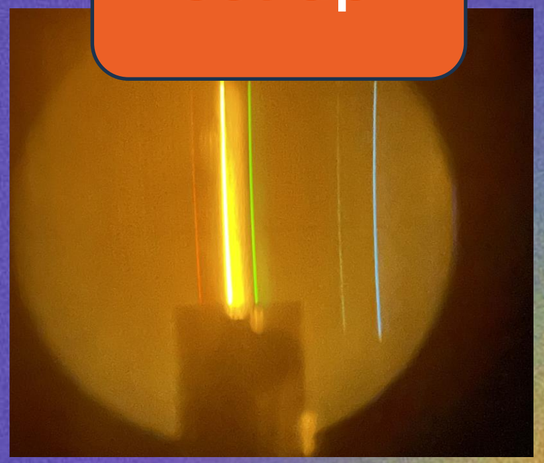
Procedure

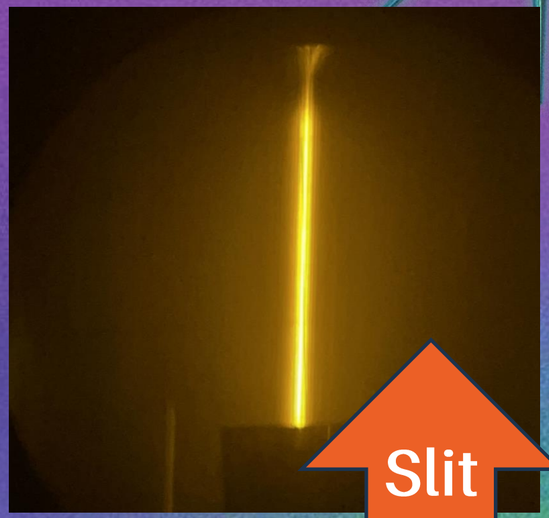
Apparatus - Demo



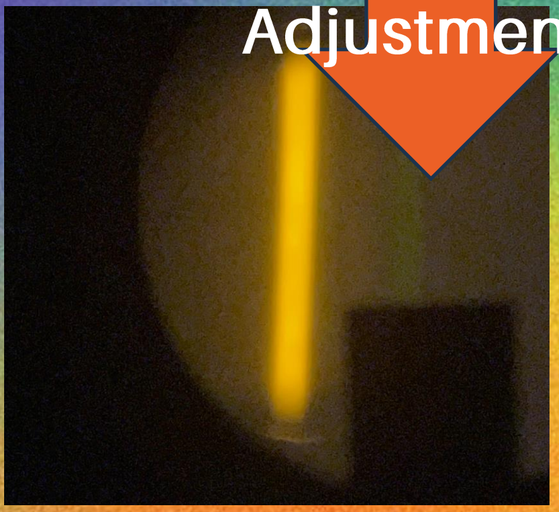


Set-up





Slit Adjustments



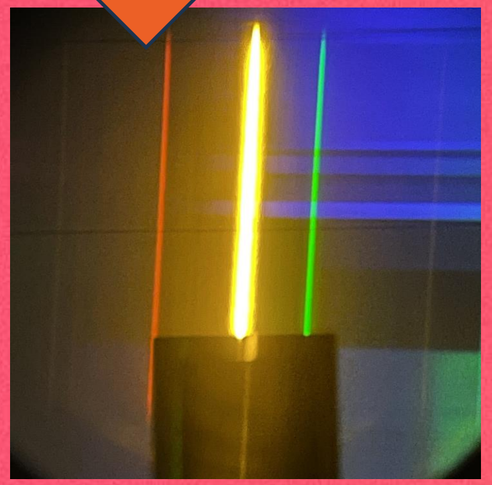
- **Spectroscope/spectrometer** -> separates a stream of light emitted by a material into visible colors/lines

Procedure

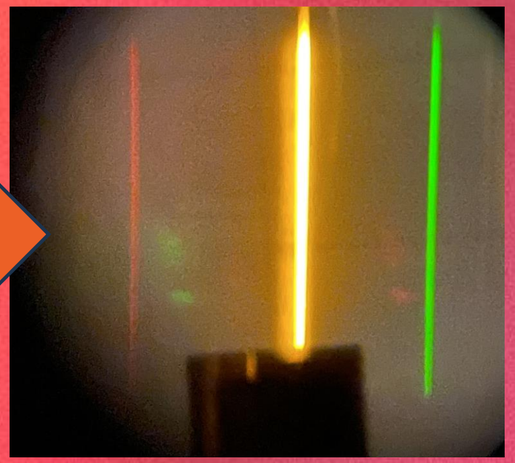
Demo: Compare what you see

Less visible spectra as order increases ->
Angular dispersion, absorption/scattering

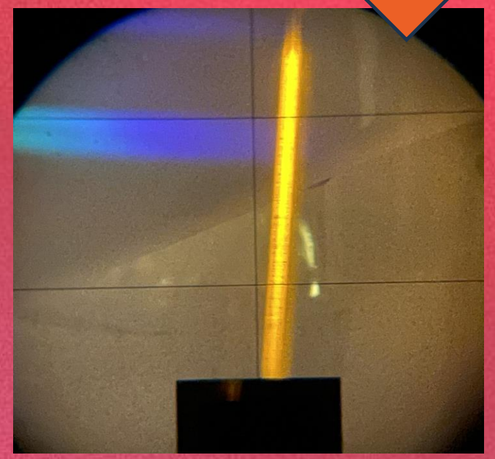
n=1

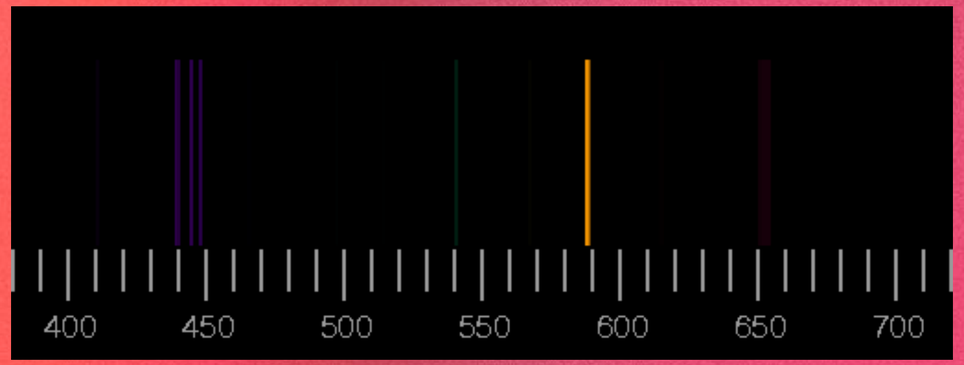


n=2



n=3

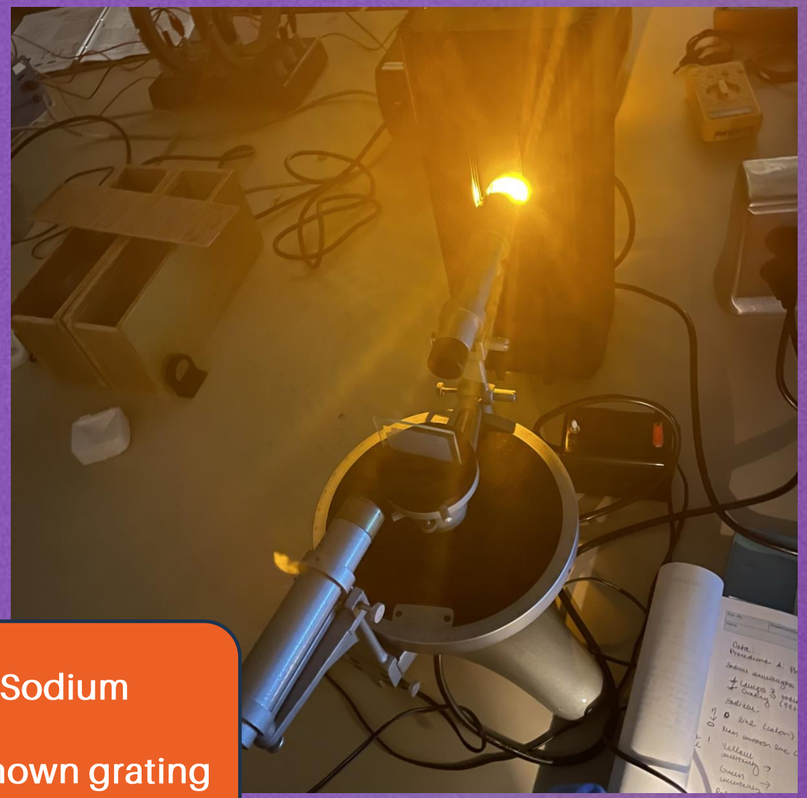




Atomic spectra (uni-hannover.de)

Procedure

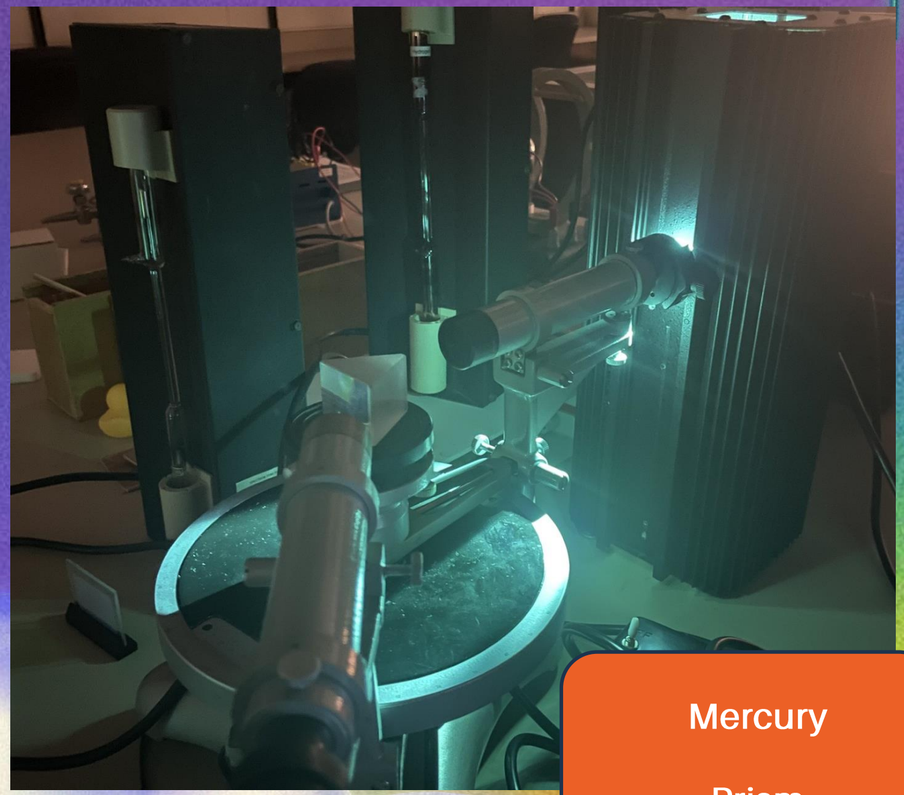
A



Sodium

Unknown grating

B



Mercury

Prism

Analysis

Organization & Conversion(s)

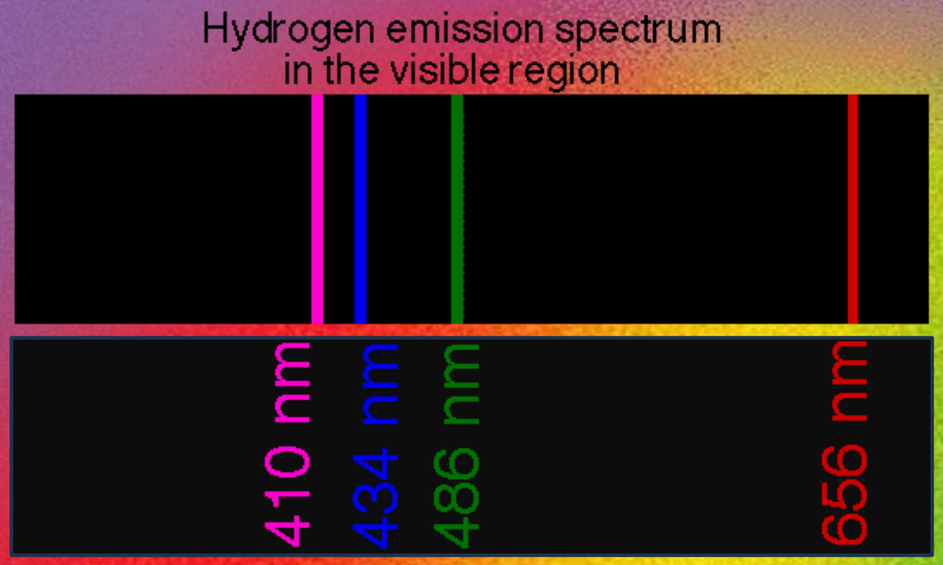
- Symmetry
- Conversions
  - Arcmin -> Degrees
  - Degrees to Radians
  - Centimeters & Nanometers

Data analysis

- Procedure A & B
  - Find ‘d’ for unknown grating
  - Brute Force Method
- A:
  - Wavelength & Rydberg
- B:
  - Wavelength, Index of Refraction

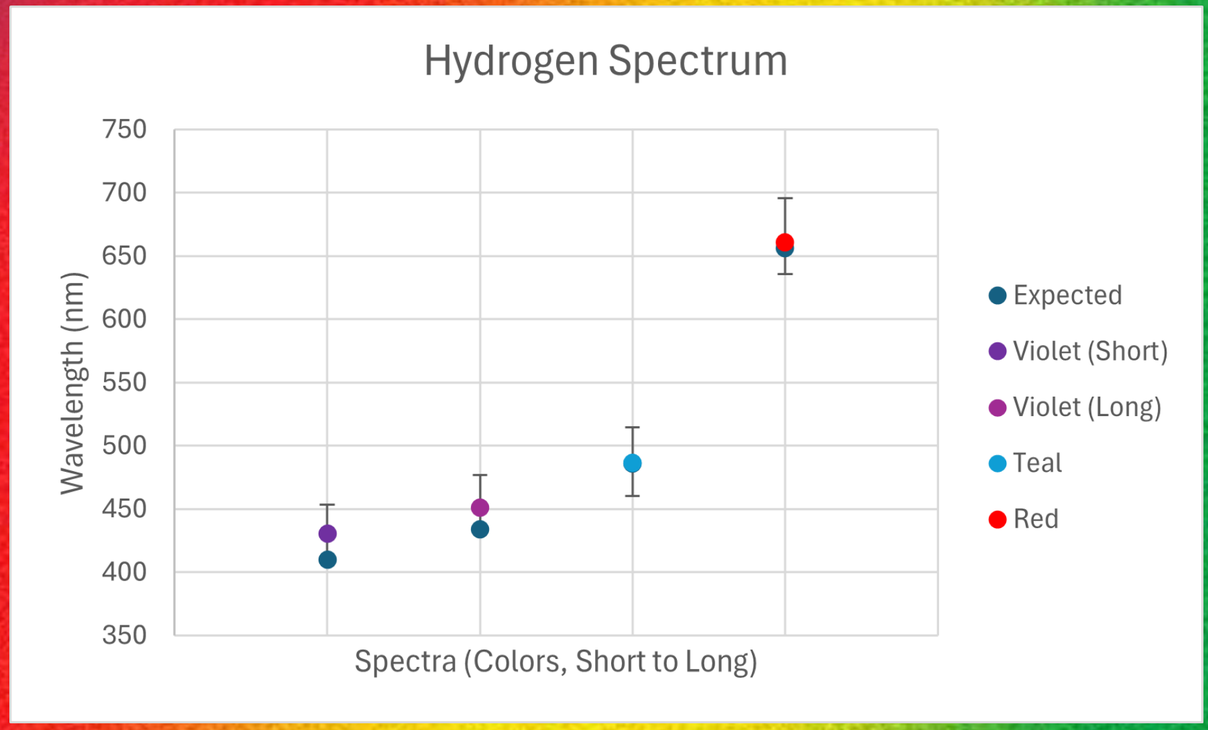
Results

Hydrogen



Activity: Hydrogen Spectrum | History of the Atomic Theory (wordpress.com)

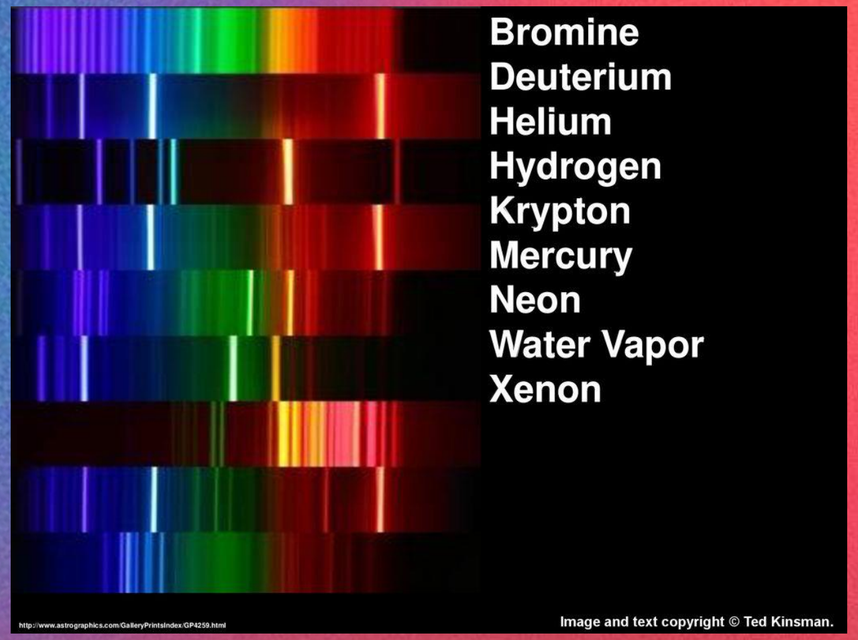
	λexpected (nm)	λbest (nm) avg	λbest (nm) STD	Percent Difference
Violet (Short)	410.17	430.53	25.26	4.84
Violet (Long)	434.05	451.32	41.66	3.9
Teal	486.13	486.29	24.5	0.34
Red	656.28	660.6	18.4	0.66

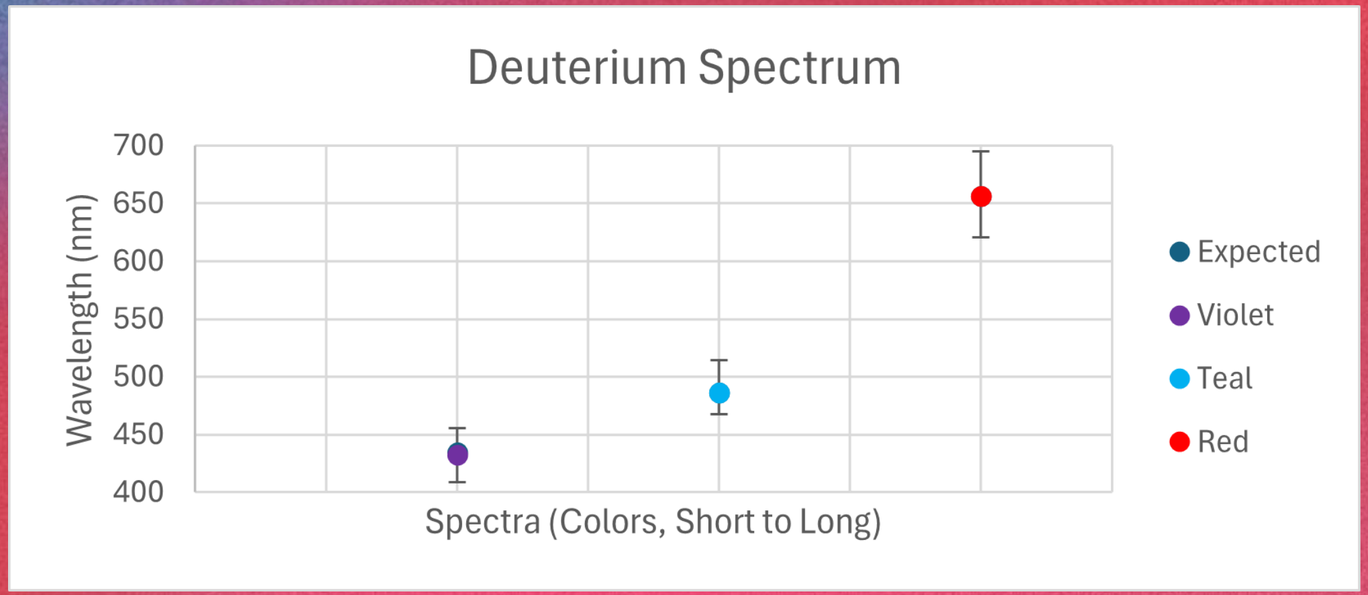


Results

Deuterium

	λexpected (nm)	λbest (nm) avg	λbest (nm) STD	Percent Difference
Violet (434.0541 nm)	434.0541	432.03	1.90	0.47
Teal	486.1407	476.54	13.24	2.00
Red	656.2899	650.14	3.54	0.94

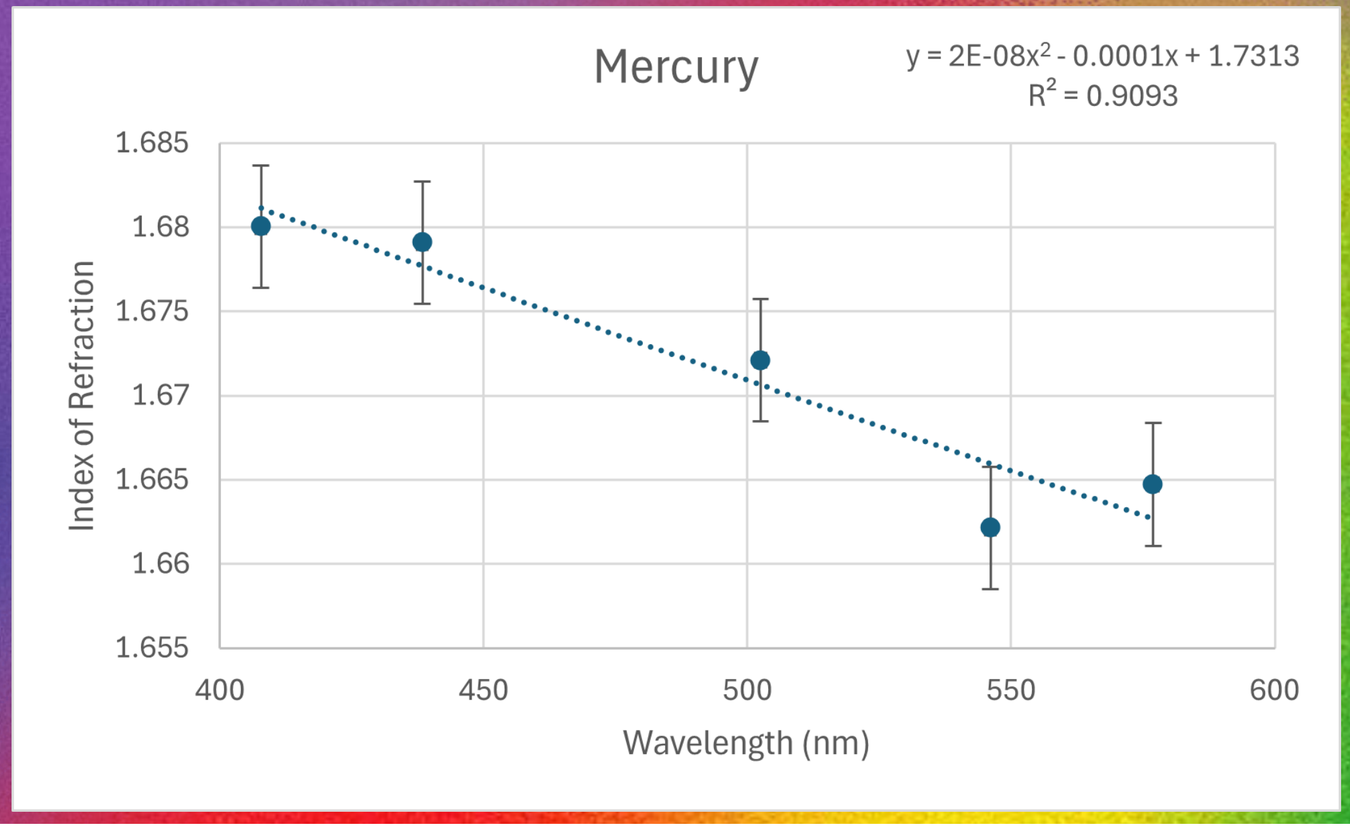




Results

Index of Refraction – Prism – Lab D

Mercury	Index of Refraction
Yellow	1.66
Green	1.66
Teal	1.67
Blue	1.67
Violet	1.68



Solids at 20°C	
Diamond	2.419
Fluorite	1.434
Glass, crown	1.52
Glass, flint	1.66

25.3: The Law of Refraction - Physics LibreTexts

Results

Rydberg Constant - Grating

	RH expected cm^-1	best (cm^-1) avg	best (cm^-1) STD	Percent Difference
Hydrogen	109737	107912.014	5374.38	1.65
Deuterium	109737	111068.028	2130.7	1.21

Balmer - Rydberg Equation

$$\frac{1}{\lambda} = R_{\infty}\left(\frac{1}{m^2} - \frac{1}{n^2}\right)$$

m *– level of a lower energy orbit*

n *– level of a higher energy orbit*

R_{∞} (R_H) is the Rydberg constant, that is equal to 1.097 x 10^{-2} nm^{-1}

[Rydberg Formula - Chemistry Steps]

Error



Mathematical

- Min/Max (Range)
- Average -> Standard Deviation
- Percent Difference
  - $\% = \left|((\Delta V)/((V_2+V_1)/2))\right| * 100$

V_1 = Initial value

V_2 = New Value

Human

- Organization
- Prism
- Measurement Methods
- Communication

Discussion & Conclusion

- How do the experimental values compare to what was expected?
  - Wavelengths
  - Rydberg constant
- Index of refraction
  - 1.66-1.67 } experimental value
    - 1.66 -> Glass, flint (25.3: The Law of Refraction - Physics LibreTexts)

Questions?

Work Cited & Links

Slide Template from SlidesCarnival

- 7.2: A Very Brief History of Spectroscopy - Physics LibreTexts
- Spectroscopy: Reading the Rainbow | HubbleSite
- 10.3: Emission Spectra - Chemistry LibreTexts
- Spectroscopy 101 – How Absorption and Emission Spectra Work | Webb (webbtelescope.org)
- 1.5: The Rydberg Formula and the Hydrogen Atomic Spectrum - Chemistry LibreTexts
- Refraction – The Physics Hypertextbook
- Harold Urey and the discovery of deuterium (ucdavis.edu)
- Sodium.ppt (uci.edu)
- Atomic Spectra (gsu.edu)
- onjhd.png (987×245) (imgur.com)
- isaac-newton-using-a-prism-to-analyze-the-colors-in-a-ray-of-light_u-L-P5ZGB10.jpg (338×450) (allpostersimages.com)
- Sir Frederick William Herschel, infrared light (IR) discoverer | NIRS Research (nirs-research.com)
- Atoms, Electrons and Photons - EnergyGroove.net
- Joseph von Fraunhofer (March 6, 1787 — June 7, 1826), German physicist, scientist | World Biographical Encyclopedia (prabook.com)